

# HAT-P-32b

R-0248

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-32\_240907 · created 2026-07-17T06:11:53+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460560.8629 +/- 0.0041 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.124 +/- 0.043                 |
| Transit depth (Rp/Rs)^2                           | 1.54 +/- 1.07 [%]               |
| Orbital Inclination (inc)                         | 89.38 +/- 1.67                  |
| Airmass coefficient 1 (a1)                        | 1.0575 +/- 0.0099               |
| Airmass coefficient 2 (a2)                        | -0.042 +/- 0.036                |
| Scatter in the residuals of the lightcurve fit is | 0.94 %                          |
| Best Comparison Star                              | #1 - [415, 332]                 |
| Optimal Aperture                                  | 4.47                            |
| Optimal Annulus                                   | 10.39                           |
| Transit Duration (day)                            | 0.1064 +/- 0.0047               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.124 ± 0.043 vs 0.1489 (deviation 0.58σ) |
| Duration fit vs published | 0.1064 d vs 0.1304 d (deviation 5.11σ)    |
| Mid-time O–C              | -10.6 min                                 |
| Depth SNR                 | 2.9                                       |
| Residual scatter          | 0.94 %                                    |

### Light curve

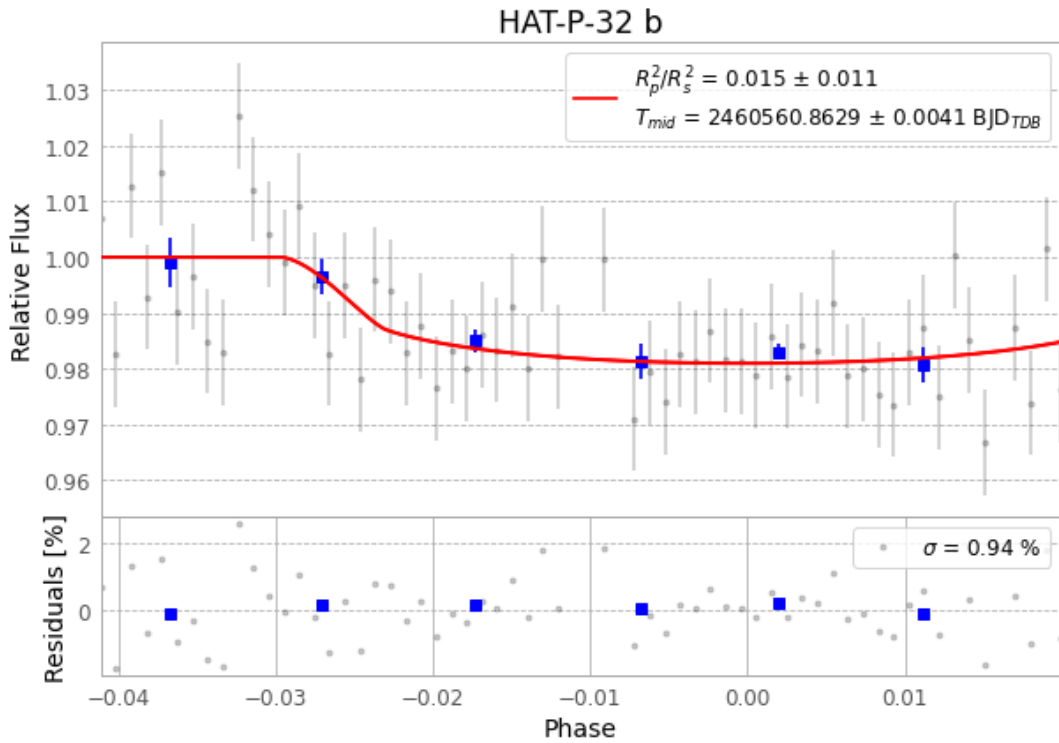


Figure 1 — FinalLightCurve\_HAT-P-32 b\_2024-09-06.png (SHA-256 0061fcb989acb268...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [343, 196], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 8dc8c61062128752...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 825eb7e

## Data provenance

64 FITS frames (42 MB), HATP-32240907063015.FITS → HATP-32240907093915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-240907.log (f980d6cc1beb...), FinalLightCurve\_HAT-P-32 b\_2024-09-06.png (0061fcb989ac...), FinalLightCurve\_HAT-P-32 b\_2024-09-06.csv (c772ee4aa30b...)

## Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 06:11Z.